

UDS Complete Reference for FCA/Stellantis Technicians

Author: Manus AI

Scope: Unified Diagnostic Services (UDS, ISO 14229 family), UDS on CAN/ISO-TP, and practical FCA/Stellantis lookup notes

Audience: Professional automotive technicians, diagnosticians, module programmers, and advanced scan-tool users

Important professional note: This document is a practical field reference, not a substitute for the paid ISO 14229 standard, FCA/Stellantis service information, wiTECH procedures, or module-specific ODX/CBF diagnostic data. UDS is standardized, but individual **DIDs, routines, security levels, seed/key algorithms, programming preconditions, and supported services vary by ECU, model year, software level, market, and network gateway configuration**. Use this reference for quick lookup and workflow orientation, then verify module-specific details in the official service tool or service information.

UDS is a request/response diagnostic protocol used by vehicle ECUs for identification, DTC handling, actuator control, routine execution, coding/configuration, and ECU programming. The protocol is defined at the application/session layers and can run over several lower-layer transports; on FCA/Stellantis CAN networks it is most commonly encountered as **UDS on CAN** using **ISO-TP / ISO 15765-2** transport framing.1 2

1. Core UDS Message Rules

UDS requests begin with a one-byte **Service Identifier (SID)**. Most positive responses use the request SID plus `0x40`, while negative responses use the standard three-byte form `7F <requested SID> <NRC>`.3 4

Item	Rule	Technician Example
Positive response SID	<code>RSID = SID + 0x40</code>	Request <code>22 F1 90</code> returns <code>62 F1 90 ...</code> for VIN read
Negative response	<code>7F <SID> <NRC></code>	<code>7F 22 31</code> means service <code>0x22</code> rejected with <code>requestOutOfRange</code>
Sub-function byte	Many services use byte 2 as sub-function	<code>10 03</code> requests extended diagnostic session

Positive response suppression	Bit 7 of many sub-function bytes may suppress a positive response	3E 80 is Tester Present with suppress-positive-response bit set
Multi-byte values	Usually big-endian / network byte order unless a DID/routine defines otherwise	DID F190 is sent as F1 90
Functional addressing	Broadcast-style request; ECUs may not respond to unsupported requests	7DF is common functional request ID on 11-bit CAN

2. SID Allocation Map: UDS Commands, Responses, Reserved Areas, and Beyond

The table below is a searchable map of the UDS SID space. It includes the commonly used services from 0x10 through 0x3E, newer services beyond 0x80, positive response ranges, and reserved or supplier-specific blocks.[5](#) [6](#)

SID / Range	Classification	Name / Meaning	Positive Response	Field Notes
0x00	Reserved	ISO reserved / not applicable	N/A	Not used as a UDS request
0x01-0x0F	OBD/J1979	Legislated OBD services	0x41-0x4F	Not UDS enhanced-diagnostics services; often available on emission ECUs
0x10	UDS request	DiagnosticSessionControl	0x50	Switches diagnostic session
0x11	UDS request	ECUReset	0x51	Requests ECU reset/power-down behavior
0x12-0x13	Reserved	ISO reserved	0x52-0x53 reserved	Do not use
0x14	UDS request	ClearDiagnosticInformation	0x54	Clears DTCs and related records

0x15-0x18	Reserved	ISO reserved	0x55-0x58 reserved	Do not use
0x19	UDS request	ReadDTCInformation	0x59	Reads DTC status, snapshot, extended data, permanent DTCs
0x1A-0x21	Reserved	ISO reserved	0x5A-0x61 reserved	Do not use
0x22	UDS request	ReadDataByIdentifier	0x62	Reads DIDs such as VIN, part number, software version
0x23	UDS request	ReadMemoryByAddress	0x63	Reads raw memory by address; usually security/session protected
0x24	UDS request	ReadScalingDataByIdentifier	0x64	Reads scaling/units/encoding for a DID if supported
0x25-0x26	Reserved	ISO reserved	0x65-0x66 reserved	Do not use
0x27	UDS request	SecurityAccess	0x67	Seed/key unlocking for protected services
0x28	UDS request	CommunicationControl	0x68	Enables/disables Tx/Rx communication types
0x29	UDS request	Authentication	0x69	Certificate/challenge-based authentication, introduced in newer UDS versions
0x2A	UDS request	ReadDataByPeriodicIdentifier	0x6A	Periodic DID transmission

0x2B	Reserved	ISO reserved	0x6B reserved	Do not use
0x2C	UDS request	DynamicallyDefineDataIdentifier	0x6C	Creates dynamic DIDs from other DIDs or memory ranges
0x2D	Reserved	ISO reserved	0x6D reserved	Do not use
0x2E	UDS request	WriteDataByIdentifier	0x6E	Writes a DID; protected on production ECUs
0x2F	UDS request	InputOutputControlByIdentifier	0x6F	Temporary actuator/signal override by DID
0x30	Reserved	ISO reserved	0x70 reserved	Do not use
0x31	UDS request	RoutineControl	0x71	Starts/stops/queries ECU routines
0x32-0x33	Reserved	ISO reserved	0x72-0x73 reserved	Do not use
0x34	UDS request	RequestDownload	0x74	Starts client-to-ECU transfer for flashing
0x35	UDS request	RequestUpload	0x75	Starts ECU-to-client transfer
0x36	UDS request	TransferData	0x76	Transfers blocks after download/upload request
0x37	UDS request	RequestTransferExit	0x77	Ends a transfer sequence
0x38	UDS request	RequestFileTransfer	0x78	File-based alternative to upload/download on ECUs that support it
0x39-0x3C	Reserved	ISO reserved	0x79-0x7C reserved	Do not use
0x3D	UDS request	WriteMemoryByAddress	0x7D	Writes raw memory by

				address; highly protected
0x3E	UDS request	TesterPresent	0x7E	Keeps non-default session/security active
0x3F-0x40	Reserved	ISO reserved / not applicable	N/A	Not normal UDS requests
0x41-0x4F	OBD/J1979 responses	Positive responses to 0x01-0x0F	N/A	Legislated OBD response range
0x50-0x7E	UDS responses	Positive responses to 0x10-0x3E	N/A	SID + 0x40 responses
0x7F	UDS response	NegativeResponse	N/A	Format 7F <SID> <NRC>
0x80-0x82	Reserved	ISO reserved / not applicable	N/A	Do not use
0x83	Withdrawn/legacy	AccessTimingParameter	0xC3	Withdrawn in ISO 14229-1:2020; may appear in older material/tools
0x84	UDS request	SecuredDataTransmission	0xC4	Wraps secured diagnostic data
0x85	UDS request	ControlDTCSetting	0xC5	Turns DTC setting on/off
0x86	UDS request	ResponseOnEvent	0xC6	Configures event-triggered responses
0x87	UDS request	LinkControl	0xC7	Requests data-link parameter changes such as baud-rate control
0x88	Reserved	ISO reserved	0xC8 reserved	Do not use
0x89-0xB9	Reserved	ISO reserved / not applicable	N/A	Do not use

0xBA-0xBE	Supplier specific	System supplier specific request services	0xFA-0xFE	Rare in aftermarket work; meaning depends on supplier
0xBF-0xC2	Reserved	ISO reserved / not applicable	N/A	Do not use
0xC3-0xC7	UDS responses	Positive responses to 0x83-0x87	N/A	Beyond-classic positive response range
0xC8	Reserved	ISO reserved	N/A	Do not use
0xC9-0xF9	Reserved	ISO reserved / not applicable	N/A	Do not use
0xFA-0xFE	Supplier specific	Positive responses to 0xBA-0xBE	N/A	Supplier-specific
0xFF	Reserved	ISO reserved / not applicable	N/A	Do not use

3. Detailed UDS Service Reference

The following table is designed for second-monitor use. The request and response formats show the UDS payload after ISO-TP reassembly, not the raw CAN frame bytes.

SID	Service	Sub-functions / Parameters	Request Format	Positive Response Format	Comments
0x10	DiagnosticSessionControl	0x01 default, 0x02 programming, 0x03 extended, 0x04 safety-system, 0x40-0x5F OEM-specific, 0x60-0x7E supplier-specific	10 <sessionType>	50 <sessionType> <P2_hi> <P2_lo> <P2*_hi> <P2*_lo>	12, 13 7E

0x11	ECUReset	0x01 hardReset, 0x02 keyOffOnReset, 0x03 softReset, 0x04 enableRapidPo werShutDown, 0x05 disableRapidPo werShutDown, OEM-specific ranges	11 <resetType>	51 <resetType> [powerDownTime]	12 , 13 33 , 7E
0x14	ClearDiagnosticInformation	No sub- function. Parameter is 3- byte groupOfDTC ; optional memory selection in newer implementations	14 <groupOfDTC_3B> [memorySelection]	54	13 , 22 33 , 72
0x19	ReadDTCInformation	DTC report type in byte 2; see full 0x19 table below	19 <reportType> <record-specific parameters>	59 <reportType> <record-specific data>	12 , 13 31 , 33
0x22	ReadDataByIdentifier	One or more 2- byte DIDs	22 <DID_hi> <DID_lo> [more DIDs]	62 <DID_hi> <DID_lo> <data> ...	13 , 14 31 , 33
0x23	ReadMemoryByAddress	Address/length format identifier plus memory address and size	23 <addrLenFormatId > <memoryAddress> <memorySize>	63 <dataRecord>	13 , 22 33 , 7F
0x24	ReadScalingDataByIdentifier	One 2-byte DID	24 <DID_hi> <DID_lo>	64 <DID_hi> <DID_lo> <scalingByte(s)>	13 , 22 33 , 7F

0x27	SecurityAccess	Odd sub-functions request seed; next even sub-functions send key. Examples: 01/02 , 03/04 , 05/06	Seed: 27 <oddLevel> ; Key: 27 <evenLevel> <key>	Seed: 67 <oddLevel> <seed> ; Key: 67 <evenLevel>	12 , 13 24 , 31 35 , 36
0x28	CommunicationControl	0x00 enableRxAndTx, 0x01 enableRxDisableTx, 0x02 disableRxEnableTx, 0x03 disableRxAndTx ; communication type byte follows	28 <controlType> <communicationType>	68 <controlType>	12 , 13 31 , 33
0x29	Authentication	0x00 deAuthenticate, 0x01 verifyCertificateUnidirectional, 0x02 verifyCertificateBidirectional, 0x03 proofOfOwnership, 0x04 transmitCertificate, 0x05 requestChallengeForAuthentication, 0x06 verifyProofOfOwnershipUnidirectional, 0x07 verifyProofOfOwnershipBidirectional, 0x08 authenticationConfiguration	29 <authenticationTask> <task-specific data>	69 <authenticationTask> <task-specific data>	12 , 13 31 , 34

0x2A	ReadDataByPeriodicIdentifier	Transmission mode plus periodic data identifiers, commonly in F2xx range	2A <transmissionMode> <periodicDID(s)>	Often 6A <transmissionMode/periodicDID> or periodic data without normal RSID depending implementation	12 , 13 31 , 33
0x2C	DynamicallyDefineDataIdentifier	0x01 defineByIdentifier, 0x02 defineByMemoryAddress, 0x03 clearDynamicallyDefinedDataIdentifier	2C <definitionType> <dynamicDID> <source definition>	6C <definitionType> <dynamicDID>	12 , 13 31 , 33
0x2E	WriteDataByIdentifier	One 2-byte DID plus data record	2E <DID_hi> <DID_lo> <dataRecord>	6E <DID_hi> <DID_lo>	13 , 22 33 , 72
0x2F	InputOutputControlByIdentifier	2-byte DID; control option commonly 0x00 returnControlToECU, 0x01 resetToDefault, 0x02 freezeCurrentState, 0x03 shortTermAdjustment	2F <DID_hi> <DID_lo> <controlOption> [controlState] [mask]	6F <DID_hi> <DID_lo> <controlOption> [status]	13 , 22 33 , 7E
0x31	RoutineControl	0x01 startRoutine, 0x02 stopRoutine, 0x03 requestRoutineResults	31 <routineControlType> <RID_hi> <RID_lo> [optionRecord]	71 <routineControlType> <RID_hi> <RID_lo> [statusRecord]	12 , 13 24 , 31 72
0x34	RequestDownload	Data format identifier, address/length format identifier,	34 <dataFormatId> <addrLenFormatId>	74 <lengthFormatId> <maxNumberOfBlockLength>	13 , 22 33 , 70

		memory address, memory size	<memoryAddress> <memorySize>		
0x35	RequestUpload	Same structure as download, but server-to-client transfer	35 <dataFormatId> <addrLenFormatId> > <memoryAddress> <memorySize>	75 <lengthFormatId> <maxNumberOfBlockLength>	13 , 22 33 , 70
0x36	TransferData	Block sequence counter plus block payload	36 <blockSequenceCounter> <transferRequestParameterRecord>	76 <blockSequenceCounter> [transferResponseParameterRecord]	13 , 24 71 , 72
0x37	RequestTransferExit	Optional transfer-exit parameter record	37 [transferRequestParameterRecord]	77 [transferResponseParameterRecord]	13 , 22 31 , 72
0x38	RequestFileTransfer	Mode of operation plus file path/name/size/compression/encryption data	38 <modeOfOperation> <filePathAndName> > <fileSize/etc.>	78 <modeOfOperation> <fileTransferParameterRecord>	13 , 22 33 , 70
0x3D	WriteMemoryByAddress	Address/length format identifier plus memory address, memory size, and data	3D <addrLenFormatId> > <memoryAddress> <memorySize> <dataRecord>	7D <addrLenFormatId> > <memoryAddress> <memorySize>	13 , 22 33 , 72
0x3E	TesterPresent	0x00 zeroSubFunction; 0x80 suppress positive response version	3E <zeroSubFunction> >	7E <zeroSubFunction> >	12 , 13
0x83	AccessTimingParameter	Legacy/withdrawn in ISO 14229-1:2020	83 <timingParameterAccessType> ...	C3 <timingParameterAccessType> ...	12 , 13 31

0x84	SecuredDataTransmission	Security wrapper for diagnostic data	84 <securityDataRequestRecord>	C4 <securityDataResponseRecord>	13 , 22 39 , 3A
0x85	ControlDTCSetting	0x01 on, 0x02 off; additional setting-control option record may be present	85 <DTCSettingType> [optionRecord]	C5 <DTCSettingType>	12 , 13 31 , 33
0x86	ResponseOnEvent	Event setup/report/stop sub-functions; implementation-specific records	86 <eventType> <eventWindowTime> <eventTypeRecord> > <serviceToRespondTo>	C6 <eventType> <eventWindowTime> > ...	12 , 13 31 , 33
0x87	LinkControl	0x01 verifyBaudrateTransitionWithFixedBaudrate, 0x02 verifyBaudrateTransitionWithSpecificBaudrate, 0x03 transitionBaudrate	87 <linkControlType> <linkControlModelIdentifier/ baud data>	C7 <linkControlType> >	12 , 13 31 , 33
0xBA-0xBE	System-supplier-specific services	Supplier-defined	<SID> <supplier data>	FA-FE <supplier data>	Depend on supplier

4. Diagnostic Session Types

Diagnostic sessions determine what functions the ECU exposes. A default session typically permits basic identification and DTC reads, while programming and extended sessions enable sensitive services only when preconditions and security are satisfied.^{3 5}

Session Type	Name	Typical Use	Technician Notes
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0x00	Reserved	Not a valid requested session	Do not request
0x01	Default Session	Normal diagnostic state after ECU boot	Basic reads usually work here; many writes/routines are locked
0x02	Programming Session	ECU flashing/reprogramming	Usually requires stable power, correct voltage, security access, and gateway authorization
0x03	Extended Diagnostic Session	Enhanced testing, actuator control, configuration, routines	Common working session for service functions and bidirectional controls
0x04	Safety System Diagnostic Session	Safety-related diagnostics	Airbag/occupant/brake/safety ECUs may support this; strict preconditions apply
0x05-0x3F	Reserved	Not standardized for normal use	Avoid unless OEM data says otherwise
0x40-0x5F	Vehicle-manufacturer-specific	FCA/Stellantis-specific sessions may appear	Meaning depends on ECU and diagnostic database
0x60-0x7E	System-supplier-specific	Supplier-specific sessions	Meaning depends on ECU supplier
0x7F	Reserved	Not valid	Do not request

A typical session entry request is `10 03` for extended diagnostic session. A positive response such as `50 03 00 32 01 F4` echoes the session and provides timing values. The `P2Server_max` value defines normal response timing, while `P2*Server_max` defines extended timing when the ECU returns `0x78 responsePending`.³

5. Security Access Flow and Levels

SecurityAccess (`0x27`) protects services that could affect immobilizer status, coding, safety functions, actuator control, or flash programming. The standard pattern is a **seed/key challenge-response**: the tester requests a seed, calculates a key using an OEM-specific algorithm, then sends the key for that level.^{1 3}

Step	Payload	Meaning	Expected Response
1	27 01	Request seed for security level 1	67 01 <seed>
2	Tester calculates key	Algorithm is OEM/module-specific	No bus message by itself
3	27 02 <key>	Send key for level 1	67 02 if accepted
4	Use protected service	Example: 31 , 2E , 34	Service-specific response

Level Pair	Common Label	Typical Use	Notes
0x01 / 0x02	Security Level 1	Enhanced diagnostic functions	Often used for routines or writable service data
0x03 / 0x04	Security Level 2	Programming or higher-risk functions	Often required before 0x34 download
0x05 / 0x06	Security Level 3	OEM/manufacturing/immobilizer functions	May be unavailable to aftermarket tools
0x07 / 0x08 and above	Additional levels	ECU/OEM-specific	Some ECUs define many odd/even pairs

Professional caution: Failed keys increment counters and may trigger lockout. Common security-related NRCs are 0x35 invalidKey , 0x36 exceedNumberOfAttempts , and 0x37 requiredTimeDelayNotExpired . Do not brute-force seed/key access on customer vehicles.

6. ReadDTCInformation (0x19) Sub-functions

Service 0x19 is the main UDS service for reading diagnostic trouble code information. It can report DTC counts, DTCs filtered by status masks, snapshot/freeze-frame records, extended data, severity, permanent DTCs, mirror memory, user-defined memory, and WWH-OBID information depending on ECU support.³

Report Type	Name	Typical Request Format	Positive Response Pattern	Technician Use
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0x01	reportNumberOfDTCByStatusMask	19 01 <statusMask>	59 01 <statusAvailabilityMask> <DTCFormatId> <DTCCount>	Count DTCs matching a status mask
0x02	reportDTCByStatusMask	19 02 <statusMask>	59 02 <statusAvailabilityMask> <DTCAndStatusRecord...>	Most common enhanced DTC read
0x03	reportDTCSnapshotIdentification	19 03	59 03 <snapshotIdentification...>	List snapshot records available
0x04	reportDTCSnapshotRecordByDTCNumber	19 04 <DTC_3B> <snapshotRecordNumber>	59 04 <DTC> <status> <snapshot data>	Read freeze-frame/snapshot for one DTC
0x05	reportDTCStoredDataByRecordNumber	19 05 <storedDataRecordNumber>	59 05 <stored data>	Read stored data by record number
0x06	reportDTCExtDataRecordByDTCNumber	19 06 <DTC_3B> <extendedDataRecordNumber>	59 06 <DTC> <status> <extended data>	Read aging counters, occurrence counters, mileage/time stamps if defined
0x07	reportNumberOfDTCBySeverityMaskRecord	19 07 <severityMask> <statusMask>	59 07 <availability> <format> <count>	Count DTCs by severity and status
0x08	reportDTCBySeverityMaskRecord	19 08 <severityMask> <statusMask>	59 08 <availability> <severity/DTC/status records>	Read DTCs by severity
0x09	reportSeverityInformationOfDTC	19 09 <DTC_3B>	59 09 <severity/status data>	Get severity for one DTC
0x0A	reportSupportedDTC	19 0A	59 0A <statusAvailabilityMask> <DTCAndStatusRecord...>	List DTCs supported by ECU

0x0B	reportFirstTestFailedDTC	19 0B	59 0B <DTC/status>	First test-failed DTC
0x0C	reportFirstConfirmedDTC	19 0C	59 0C <DTC/status>	First confirmed DTC
0x0D	reportMostRecentTestFailedDTC	19 0D	59 0D <DTC/status>	Most recent test-failed DTC
0x0E	reportMostRecentConfirmedDTC	19 0E	59 0E <DTC/status>	Most recent confirmed DTC
0x0F	reportMirrorMemoryDTCByStatusMask	19 0F <statusMask>	59 0F ...	Mirror memory; withdrawn in newer UDS but may be seen in older systems
0x10	reportMirrorMemoryDTCExtDataRecordByDTCNumber	19 10 <DTC_3B> <record>	59 10 ...	Mirror-memory extended data; legacy/withdrawn in newer UDS
0x11	reportNumberOfMirrorMemoryDTCByStatusMask	19 11 <statusMask>	59 11 ...	Count mirror-memory DTCs
0x12	reportNumberOfEmissionsOBDDTCByStatusMask	19 12 <statusMask>	59 12 ...	Count emissions OBD DTCs
0x13	reportEmissionsOBDDTCByStatusMask	19 13 <statusMask>	59 13 ...	Read emissions OBD DTCs through UDS
0x14	reportDTCFaultDetectionCounter	19 14	59 14 <DTC/faultDetectionCounter...>	Read fault detection counters
0x15	reportDTCWithPermanentStatus	19 15	59 15 <DTC/status...>	Permanent DTC list where supported
0x16	reportDTCExtDataRecordByRecordNumber	19 16 <recordNumber>	59 16 ...	Extended data by record number
0x17	reportUserDefinedMemoryDTCByStatusMask	19 17 <statusMask> <memorySelection>	59 17 ...	User-defined DTC memory

0x18	reportUserDefMemoryDTCSnapshotRecordByDTCNumber	19 18 <DTC_3B> <snapshotRecord> <memorySelection>	59 18 ...	Snapshot from selected user-defined memory
0x19	reportUserDefMemoryDTCExtDataRecordByDTCNumber	19 19 <DTC_3B> <record> <memorySelection>	59 19 ...	Extended data from selected user-defined memory
0x1A	reportSupportedDTCExtDataRecord	19 1A <DTC_3B>	59 1A ...	Supported extended data records for a DTC
0x42	reportWWHOBDDTCByMaskRecord	19 42 <functionalGroupId> <statusMask> <severityMask>	59 42 ...	WWH-OBDDTC filtering
0x55	reportWWHOBDDTCWithPermanentStatus	19 55 <functionalGroupId>	59 55 ...	WWH-OBDDTC permanent DTCs
0x56	reportDTCInformationByDTCReadinessGroupIdIdentifier	19 56 <readinessGroupId>	59 56 ...	DTC/readiness group information

DTC Status Byte Quick Lookup

UDS DTC records usually include a status byte. The exact status availability mask is reported by the ECU, but the conventional bit meanings are as follows.

Bit	Mask	Name	Field Meaning
0	0x01	testFailed	Test currently failed
1	0x02	testFailedThisOperationCycle	Failed in current operation cycle
2	0x04	pendingDTC	Pending fault
3	0x08	confirmedDTC	Confirmed/stored fault
4	0x10	testNotCompletedSinceLastClear	Monitor has not completed since clear

5	0x20	testFailedSinceLastClear	Failed at least once since clear
6	0x40	testNotCompletedThisOperationCycle	Monitor has not completed this cycle
7	0x80	warningIndicatorRequested	MIL/warning indicator requested

7. RoutineControl (0x31) Sub-functions and Practical Uses

RoutineControl executes ECU-defined routines. A routine is selected by a two-byte **Routine Identifier (RID)**, and the meaning of RIDs is OEM/module-specific. Common programming-related routines include memory erase, checksum/CRC verification, programming dependency checks, and application validation, but the numeric RIDs vary by ECU.[7](#)

Sub-function	Name	Request Format	Positive Response	Common Uses
0x01	startRoutine	31 01 <RID_hi> <RID_lo> [optionRecord]	71 01 <RID_hi> <RID_lo> [statusRecord]	Start self-test, erase, calibration routine, actuator routine, programming check
0x02	stopRoutine	31 02 <RID_hi> <RID_lo> [optionRecord]	71 02 <RID_hi> <RID_lo> [statusRecord]	Stop a running test/routine if supported
0x03	requestRoutineResults	31 03 <RID_hi> <RID_lo> [optionRecord]	71 03 <RID_hi> <RID_lo> [statusRecord]	Read completion/result status

NRC	Why It Appears During RoutineControl
0x12	RoutineControl sub-function is not supported
0x13	Wrong length or missing RID/option record

0x22	Preconditions are not met; example: vehicle speed, gear range, voltage, temperature, engine state
0x24	Wrong sequence; example: requesting results before start, starting a routine already active
0x31	RID not supported, not available in current session, or option record out of range
0x33	Security access required
0x72	Routine failed during memory/programming operation

8. Data Transfer and Flashing Services (0x34 , 0x36 , 0x37)

UDS flashing is normally a controlled sequence, not just three services in isolation. On FCA/Stellantis vehicles, the gateway, ECU, battery voltage, immobilizer state, software part compatibility, and session/security state all matter. The high-level UDS pattern is standard, but the exact data blocks and signatures are OEM-specific.

Phase	Typical UDS Payload	Purpose	Technician Notes
1	10 02	Enter programming session	ECU may reset or stop normal communication
2	27 <seed level> then 27 <key level> <key>	Unlock programming security	Seed/key is OEM-specific
3	28 ... and/or 85 02	Optional communication/DTC-control setup	Used to reduce bus chatter and prevent DTC setting during programming
4	31 01 <erase RID> ...	Erase memory routine	Often returns 0x78 responsePending while erase runs
5	34 <DFI> <ALFI> <address> <size>	Request download	ECU returns maximum block length

6	36 <BSC> <data> repeated	Transfer data blocks	Block sequence counter rolls from 0x01 upward and wraps after 0xFF
7	37 [params]	Transfer exit	Ends the transfer; ECU may validate transfer framing
8	31 01 <verify/check RID> ...	Verify checksum, signature, dependencies	Required before booting new software
9	11 01 or 11 03	Reset ECU	ECU restarts into application if programming succeeded
10	85 01 / restore communication if needed	Return DTC setting/communication to normal	Some ECUs restore automatically after reset

Service	Key Fields	What to Watch
0x34 RequestDownload	dataFormatIdentifier defines compression/encryption; addressAndLengthFormatIdentifier defines byte counts for address and size	0x70 uploadDownloadNotAccepted often means wrong session, security, address, size, signature, or preconditions
0x36 TransferData	blockSequenceCounter plus data	0x73 wrongBlockSequenceCounter means the ECU detected an unexpected block counter; repeating the last accepted block may be allowed depending on ECU
0x37 RequestTransferExit	Optional final transfer parameters	0x24 requestSequenceError means download/upload sequence was not properly active or was already aborted

9. CommunicationControl (0x28) and ControlDTCSetting (0x85)

These two services are frequently encountered during programming and guided diagnostic procedures. They are powerful because they alter ECU behavior while diagnostics are active.

CommunicationControl (0x28)

Control Type	Name	Effect
0x00	enableRxAndTx	Enable receiving and transmitting selected communication
0x01	enableRxAndDisableTx	ECU receives selected communication but does not transmit it
0x02	disableRxAndEnableTx	ECU transmits selected communication but ignores selected received communication
0x03	disableRxAndTx	Disable both receiving and transmitting selected communication
0x04-0x3F	Reserved	Not standardized
0x40-0x5F	Vehicle-manufacturer-specific	FCA/Stellantis meanings require ECU data
0x60-0x7E	Supplier-specific	Supplier-defined

The `communicationType` byte selects what kind of communication is affected. OEMs may separate normal communication, network-management communication, and application-specific communication. Incorrect use can temporarily silence a module or create network DTCs.

ControlDTCSetting (0x85)

DTC Setting Type	Name	Request	Positive Response	Practical Use
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0x01	on	85 01 [optionRecord]	C5 01	Restore DTC setting
0x02	off	85 02 [optionRecord]	C5 02	Prevent new DTC logging during programming or controlled tests
0x03-0x3F	Reserved	N/A	N/A	Do not use unless OEM data says otherwise
0x40-0x5F	Vehicle-manufacturer-specific	OEM-defined	OEM-defined	FCA/Stellantis-specific behavior depends on ECU
0x60-0x7E	Supplier-specific	Supplier-defined	Supplier-defined	ECU supplier-specific

10. Common Data Identifiers (DIDs), Including FCA/Stellantis-Oriented Identification DIDs

A UDS **Data Identifier (DID)** is a two-byte identifier used with services such as 0x22 ReadDataByIdentifier , 0x2E WriteDataByIdentifier , 0x2F InputOutputControlByIdentifier , 0x2A ReadDataByPeriodicIdentifier , and 0x2C DynamicallyDefineDataIdentifier .⁸ The F180-F19F area is especially useful for ECU and vehicle identification.

DID	Standard Name / Common Meaning	Typical FCA/Stellantis Technician Use	Example Request
F180	Boot Software Identification	Bootloader ID/number	22 F1 80
F181	Application Software Identification	Application software identification	22 F1 81
F182	Application Data Identification	Calibration/data package identification	22 F1 82
F183	Boot Software Fingerprint	Last bootloader programming fingerprint	22 F1 83
F184	Application Software Fingerprint	Last application programming fingerprint	22 F1 84

F185	Application Data Fingerprint	Last calibration/data programming fingerprint	22 F1 85
F186	Active Diagnostic Session	Current diagnostic session	22 F1 86
F187	Vehicle Manufacturer Spare Part Number	Common FCA/Stellantis part number / ECU spare part number lookup	22 F1 87
F188	Vehicle Manufacturer ECU Software Number	ECU software number	22 F1 88
F189	Vehicle Manufacturer ECU Software Version Number	Software version / flash level lookup	22 F1 89
F18A	System Supplier Identifier	Supplier code/name	22 F1 8A
F18B	ECU Manufacturing Date	ECU build date	22 F1 8B
F18C	ECU Serial Number	ECU serial number; useful for module replacement records	22 F1 8C
F18D	Supported Functional Units	Functional units implemented by ECU	22 F1 8D
F18E	Vehicle Manufacturer Kit/Assembly Part Number	Assembly/kit part number; may align with service part data	22 F1 8E
F18F	Regulation X Software Identification Numbers	Regulatory software ID where required	22 F1 8F
F190	VIN Data Identifier	Vehicle Identification Number; 17 ASCII characters when supported	22 F1 90
F191	Vehicle Manufacturer ECU Hardware Number	ECU hardware number / HW part number	22 F1 91
F192	System Supplier ECU Hardware Number	Supplier hardware number	22 F1 92
F193	System Supplier ECU Hardware Version Number	Supplier hardware version	22 F1 93

F194	System Supplier ECU Software Number	Supplier software number	22 F1 94
F195	System Supplier ECU Software Version Number	Supplier software version	22 F1 95
F196	Exhaust Regulation / Type Approval Number	Emissions/type approval reference	22 F1 96
F197	System Name or Engine Type	System name, engine type, or ECU family	22 F1 97
F198	Repair Shop Code / Tester Serial Number	Tester/shop code from last programming	22 F1 98
F199	Programming Date	Last programming date	22 F1 99
F19A	Calibration Repair Shop Code / Equipment Serial	Calibration shop/tester identifier	22 F1 9A
F19B	Calibration Date	Last calibration date	22 F1 9B
F19C	Calibration Equipment Software Number	Programming/calibration tool software number	22 F1 9C
F19D	ECU Installation Date	ECU installation date if stored	22 F1 9D
F19E	ODX File Identifier	Diagnostic database reference	22 F1 9E
F19F	Entity Identifier	Entity reference for secured data transmission	22 F1 9F
FF00	UDS Version Data Identifier	UDS version implemented by server	22 FF 00
FF01	ISO 15765-5 / CAN capability indicator	CAN Classical/CAN FD support where implemented	22 FF 01

DID Ranges Useful for Search and Troubleshooting

DID Range	Allocation	Practical Meaning
0000-00FF	ISO/SAE reserved	Do not assume support

0100-A5FF	Vehicle-manufacturer-specific	FCA/Stellantis enhanced data may live here
A600-A7FF	Legislative reserved	Regulatory use
A800-ACFF	Vehicle-manufacturer-specific	OEM-specific data
AD00-AFFF	Legislative reserved	Regulatory use
B000-B1FF	Vehicle-manufacturer-specific	OEM-specific data
B200-BFFF	Legislative reserved	Regulatory use
C000-C2FF	Vehicle-manufacturer-specific	OEM-specific data
C300-CEFF	Legislative reserved	Regulatory use
CF00-EFFF	Vehicle-manufacturer-specific	OEM-specific data
F010-F0FF	Vehicle-manufacturer-specific	OEM identification/configuration
F100-F17F	Vehicle-manufacturer identification options	Often OEM-specific identification
F180-F19F	Standardized identification DIDs	Best first scan for ECU identity
F1A0-F1EF	Vehicle-manufacturer identification options	OEM-specific IDs; often useful on FCA/Stellantis but database-dependent
F1F0-F1FF	System-supplier identification options	Supplier-specific IDs
F200-F2FF	Periodic Data Identifiers	Used with 0x2A
F300-F3FF	Dynamically Defined Data Identifiers	Used with 0x2C
F400-F8FF	OBD-related DID areas	Regulated OBD data/info/monitoring values
F900-F9FF	Tachograph DIDs	Heavy/commercial applications
FA00-FAFF	Safety/EDR-related DIDs	Airbag/EDR/safety system data where supported
FD00-FEFF	System-supplier-specific	Supplier-defined
FF00-FFFF	UDS version / reserved	FF00 and FF01 are useful when supported

FCA/Stellantis-Oriented DID Notes

For FCA/Stellantis work, the most consistently useful public/standardized DID requests are **F190** for VIN, **F187** for ECU spare part number, **F189** for software version, **F18C** for ECU serial number, and **F191** for ECU hardware number. Many modules also expose additional part, supplier, calibration, and boot/application identifiers around **F180-F19F**, while other DIDs in manufacturer-specific ranges require wiTECH, ODX data, or module-specific service information.

Technician Question	First DIDs to Try	Interpretation Notes
“What VIN is stored in this module?”	F190	Expect 17 ASCII VIN bytes if supported; some modules may not store VIN or may require session/security
“What part number is this ECU reporting?”	F187 , F18E , F191 , F192	Compare spare part, assembly, VM hardware, and supplier hardware numbers
“What flash/software level is installed?”	F188 , F189 , F181 , F182 , F194 , F195	FCA/Stellantis service tools may display these with OEM formatting
“Was this ECU programmed recently?”	F183-F185 , F198 , F199 , F19A , F19B , F19C	Fingerprint/date/shop-code support is ECU-dependent
“What supplier or system family is this?”	F18A , F197 , F19E	Useful when comparing modules across platforms/model years

11. Negative Response Codes (NRC) Complete Practical Table

A negative UDS response has the format **7F <requestedSID> <NRC>**. For example, **7F 31 22** means a RoutineControl request was rejected because conditions were not correct.[3](#) [4](#)

NRC	Name	Description / Technician Interpretation
0x00	positiveResponse	Reserved for internal server use; not used in an actual

		negative response
0x01-0x0F	ISO/SAE reserved	Reserved range
0x10	generalReject	ECU rejected the request and no more specific NRC applies
0x11	serviceNotSupported	ECU does not support the requested SID
0x12	subFunctionNotSupported	SID exists, but requested sub-function is unsupported
0x13	incorrectMessageLengthOrInvalidFormat	Payload length or parameter format is wrong
0x14	responseTooLong	Positive response would exceed transport/server buffer capability
0x15-0x20	ISO reserved	Reserved range
0x21	busyRepeatRequest	ECU is temporarily busy; repeat the request later
0x22	conditionsNotCorrect	Preconditions are not met; check voltage, speed, gear, engine state, temperature, session, routine state
0x23	ISO reserved	Reserved
0x24	requestSequenceError	Request order is wrong; common in security, routines, and transfer sequences
0x25	noResponseFromSubnetComponent	Gateway/server could not get required response from another ECU/subnet
0x26	failurePreventsExecutionOfRequestedAction	A DTC/failure condition prevents the action
0x27-0x30	ISO reserved	Reserved range
0x31	requestOutOfRange	Parameter, DID, RID, memory address, or option is unsupported/out of range or unavailable in active session
0x32	ISO reserved	Reserved

0x33	securityAccessDenied	Security state is not unlocked or access rights are insufficient
0x34	authenticationRequired	Authentication state is insufficient for the requested service
0x35	invalidKey	SecurityAccess key did not match the ECU calculation
0x36	exceedNumberOfAttempts	Too many invalid security attempts; lockout active
0x37	requiredTimeDelayNotExpired	Security delay timer has not expired
0x38	secureDataTransmissionRequired	Service must be sent via secured diagnostic channel
0x39	secureDataTransmissionNotAllowed	Request was sent through secured wrapper but is not allowed that way
0x3A	secureDataVerificationFailed	Secure data verification failed
0x3B-0x4F	ISO reserved	Reserved range
0x50	certificateVerificationFailedInvalidTimePeriod	Certificate validity time period is invalid
0x51	certificateVerificationFailedInvalidSignature	Certificate signature could not be verified
0x52	certificateVerificationFailedInvalidChainOfTrust	Chain of trust is invalid
0x53	certificateVerificationFailedInvalidType	Certificate type is not valid for this use
0x54	certificateVerificationFailedInvalidFormat	Certificate format is invalid
0x55	certificateVerificationFailedInvalidContent	Certificate content does not match requirements
0x56	certificateVerificationFailedInvalidScope	Certificate scope does not match requested access
0x57	certificateVerificationFailedInvalidCertificate	Certificate is invalid or revoked
0x58	ownershipVerificationFailed	Ownership proof failed

0x59	challengeCalculationFailed	ECU could not calculate challenge
0x5A	settingAccessRightsFailed	ECU could not set requested access rights
0x5B	sessionKeyCreationDerivationFailed	Session key creation/derivation failed
0x5C	configurationDataUsageFailed	Configuration data could not be used
0x5D	deAuthenticationFailed	De-authentication failed
0x5E-0x6F	ISO reserved	Reserved range
0x70	uploadDownloadNotAccepted	Download/upload request rejected; common during flashing setup problems
0x71	transferDataSuspended	Active transfer was suspended due to fault
0x72	generalProgrammingFailure	ECU detected programming/erase/write failure in nonvolatile memory
0x73	wrongBlockSequenceCounter	TransferData block sequence counter is wrong
0x74-0x77	ISO reserved	Reserved range
0x78	requestCorrectlyReceivedResponsePending	ECU accepted request but needs more time; wait for final response
0x79-0x7D	ISO reserved	Reserved range
0x7E	subFunctionNotSupportedInActiveSession	Change diagnostic session or use another sub-function
0x7F	serviceNotSupportedInActiveSession	Service exists but is not available in active session
0x80	ISO reserved / conditions range base	Reserved in many references; 0x80-0xFF is the conditions-not-correct specific range concept
0x81	rpmTooHigh	Engine speed too high for requested action

0x82	rpmTooLow	Engine speed too low
0x83	enginelRunning	Engine must not be running
0x84	enginelNotRunning	Engine must be running
0x85	engineRunTimeTooLow	Engine has not run long enough
0x86	temperatureTooHigh	Temperature too high
0x87	temperatureTooLow	Temperature too low
0x88	vehicleSpeedTooHigh	Vehicle speed too high
0x89	vehicleSpeedTooLow	Vehicle speed too low
0x8A	throttlePedalTooHigh	Throttle/pedal position too high
0x8B	throttlePedalTooLow	Throttle/pedal position too low
0x8C	transmissionRangeNotInNeutral	Transmission not in neutral
0x8D	transmissionRangeNotInGear	Transmission not in gear
0x8E	ISO reserved	Reserved / not commonly assigned
0x8F	brakeSwitchNotClosed	Brake pedal/switch condition not satisfied
0x90	shifterLeverNotInPark	Shifter not in Park
0x91	torqueConverterClutchLocked	Torque converter clutch locked
0x92	voltageTooHigh	System voltage too high
0x93	voltageTooLow	System voltage too low
0x94	resourceTemporarilyNotAvailable	Required resource is temporarily unavailable
0x95-0xEF	Reserved / vehicle-specific conditions	Treat as OEM-specific unless documented
0xF0-0xFE	Vehicle-manufacturer-specific conditions	FCA/Stellantis-specific meanings require OEM diagnostic data
0xFF	Reserved	Reserved

12. ISO-TP / ISO 15765-2 Framing Basics for UDS on CAN

Classic CAN frames carry only up to eight data bytes. UDS payloads may be longer, so ISO-TP segments and reassembles diagnostic payloads. The **Protocol Control Information (PCI)** nibble in the first data byte identifies the ISO-TP frame type.^{1 2}

Frame Type	PCI High Nibble	Name	Purpose	Classic CAN Data Example
0	0x0	Single Frame (SF)	Carries a complete payload up to 7 bytes on normal addressing	03 22 F1 90 00 00 00 00 means 3-byte UDS payload 22 F1 90
1	0x1	First Frame (FF)	Starts a multi-frame message and includes 12-bit payload length	10 14 62 F1 90 ... starts a 20-byte response
2	0x2	Consecutive Frame (CF)	Continues a multi-frame transfer; low nibble is sequence number	21 ... , 22 ... , up to 2F , then wraps to 20
3	0x3	Flow Control (FC)	Receiver controls sender pacing	30 00 00 means ClearToSend, block size 0, STmin 0

ISO-TP Flow Control Bytes

Byte	Field	Meaning
1	Flow Status	0x30 ClearToSend, 0x31 Wait, 0x32 Overflow/Abort

2	Block Size	Number of consecutive frames allowed before next flow control; 0x00 often means unlimited for that message
3	STmin	Minimum separation time between consecutive frames; values can encode milliseconds or microsecond ranges depending value

Raw CAN Example: Reading VIN F190

Direction	CAN ID Example	CAN Data Bytes	Decoded Meaning
Tester to ECU	7E0	03 22 F1 90 00 00 00 00	Single-frame request: UDS payload 22 F1 90
ECU to Tester	7E8	10 14 62 F1 90 31 43 34	First frame: total UDS payload length 0x014 = 20 bytes; begins 62 F1 90 1C4...
Tester to ECU	7E0	30 00 00 00 00 00 00 00	Flow control ClearToSend
ECU to Tester	7E8	21 52 4A 42 41 42 43 44	Consecutive frame sequence 1
ECU to Tester	7E8	22 45 46 47 48 31 32 33	Consecutive frame sequence 2; remaining VIN bytes

13. CAN Addressing for FCA/Stellantis and Common UDS-on-CAN Work

On 11-bit CAN diagnostics, many vehicles follow the common ISO-TP/OBD diagnostic addressing pattern: functional request 0x7DF, physical requests in the 0x7E0-0x7E7 range, and physical responses in the 0x7E8-0x7EF range.⁹ The user-requested FCA/Stellantis quick rule is shown below.

CAN ID	Direction / Type	Typical Meaning	Notes
0x7DF	Tester to vehicle, functional	Functional broadcast request	Multiple ECUs may receive; often used for OBD and simple functional UDS requests
0x7E0	Tester to ECU, physical	Common PCM/ECM request ID	Response often 0x7E8
0x7E1	Tester to ECU, physical	Common TCM or next powertrain ECU request ID	Response often 0x7E9
0x7E2-0x7E7	Tester to ECU, physical	Additional physical request IDs	Exact ECU mapping varies
0x7E8	ECU to tester, physical response	Response to 0x7E0	Normal response offset is often +0x08
0x7E9	ECU to tester, physical response	Response to 0x7E1	Normal response offset is often +0x08
0x7EA-0x7EF	ECU to tester, physical responses	Additional ECU response IDs	Exact ECU mapping varies
0x7E0-0x7EF	Practical shorthand	Physical diagnostic range	Field tools often describe this entire block as diagnostic physical addressing

Addressing Cautions on FCA/Stellantis Vehicles

Modern FCA/Stellantis vehicles may route diagnostics through a gateway and may use multiple CAN buses, CAN FD, Ethernet diagnostics, gateway authentication, or security gateways depending on model year and platform. A module that does not respond on 7E0-7EF is not necessarily offline; it may be behind a gateway, on another network, require a different addressing format, require functional addressing first, or require an authenticated scan tool path.

14. Fast Technician Lookup: Common Requests

Task	UDS Payload	Expected Positive Response	Notes
Enter extended session	10 03	50 03 ...	Often prerequisite for active tests/routines
Enter programming session	10 02	50 02 ...	Use only during proper programming procedure
Keep session alive	3E 00	7E 00	Send periodically before S3 timeout; many tools use suppress-positive 3E 80
Read VIN	22 F1 90	62 F1 90 <VIN>	Response usually multi-frame because VIN response is 20 payload bytes
Read ECU part number	22 F1 87	62 F1 87 ...	Format is ECU/OEM-specific
Read software version	22 F1 89	62 F1 89 ...	Useful for flash-level confirmation
Read hardware number	22 F1 91	62 F1 91 ...	Useful for replacement-module comparison
Read ECU serial number	22 F1 8C	62 F1 8C ...	Useful for documentation
Read active DTCs by status mask	19 02 <mask>	59 02 ...	0xFF mask asks for all status bits supported by many ECUs
Clear all DTCs	14 FF FF FF	54	Some ECUs require ignition/session/security/preconditions
Request SecurityAccess seed level 1	27 01	67 01 <seed>	Algorithm is OEM/module-specific
Send SecurityAccess key level 1	27 02 <key>	67 02	Bad key may trigger lockout
Start routine	31 01 <RID>	71 01 <RID> ...	RID is ECU-specific

Turn DTC setting off	85 02	C5 02	Often used during programming; restore afterward
Turn DTC setting on	85 01	C5 01	Restores normal DTC logging

15. Troubleshooting by Negative Response

Symptom	Likely NRC	What to Check First
ECU says service is not available	0x11 or 0x7F	Wrong ECU, unsupported service, wrong session
ECU says sub-function not supported	0x12 or 0x7E	Wrong session, wrong sub-function, unsupported control type
ECU rejects message immediately	0x13	Payload length, missing DID/RID bytes, ISO-TP padding/addressing
DID or RID does not work	0x31	DID/RID not supported, wrong session, wrong security state, wrong ECU
Routine will not start	0x22	Voltage, engine state, gear/park/neutral, brake switch, temperature, vehicle speed, DTCs
Security unlock fails	0x35 , 0x36 , 0x37	Wrong key/level, lockout timer, too many attempts
Flash download rejected	0x70	Wrong session/security, address/size, file compatibility, gateway authorization, voltage
Transfer stops mid-flash	0x71 , 0x72 , 0x73	Bus errors, wrong block counter, memory failure, wrong block size, tool timing
ECU keeps returning pending	0x78	Wait for final response; do not send a different request unless procedure says to

Preconditions-specific reject

0x81-0x94

Follow the named condition exactly, especially RPM, engine state, brake, Park/Neutral, and voltage

16. References

Footnotes

1. [CSS Electronics, “UDS Explained - A Simple Intro \(Unified Diagnostic Services\)”](#) . ↩ ↩2 ↩3
2. [RAMN Documentation, “Interacting with UDS”](#) . ↩ ↩2
3. [py-uds Documentation, “Diagnostic Services”](#) . ↩ ↩2 ↩3 ↩4 ↩5 ↩6
4. [py-uds Documentation, “Diagnostic Message”](#) . ↩ ↩2
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6. [ISO Online Browsing Platform, ISO 14229-1:2020 listing](#). ↩
7. [EmbeTronicX, “Remote Activation of Routine Control in UDS Protocol”](#) . ↩
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9. [Wikipedia, “Unified Diagnostic Services”](#) . ↩